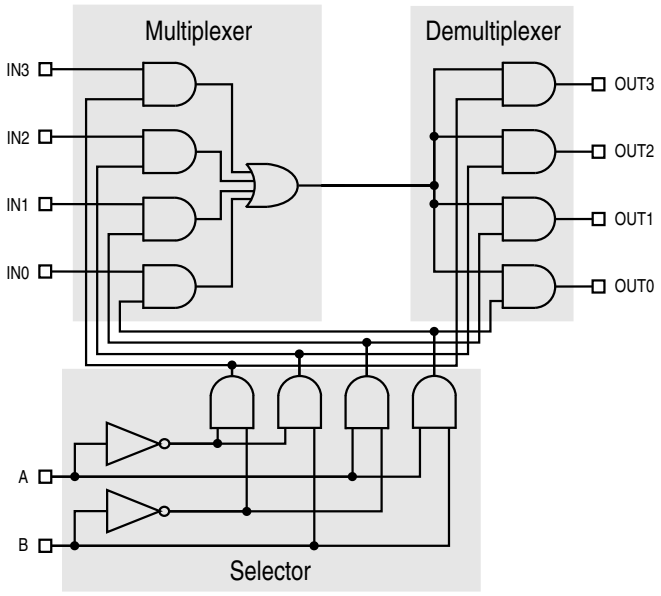
Systems Architecture

The Tri-State, Bus and Memory

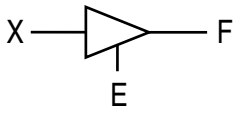
Signal Driving

- Signal shared between many devices
- Each device drains the input signal
- **Fan Out**
  The number of devices an output signal can drive before is too week to use
- Fan Out is 4
- Use Switches to over this limitation
- **Multiplexer**
  Select input device
- **Demultiplexer**
  Select output device

Multiplexer / Demultiplexer



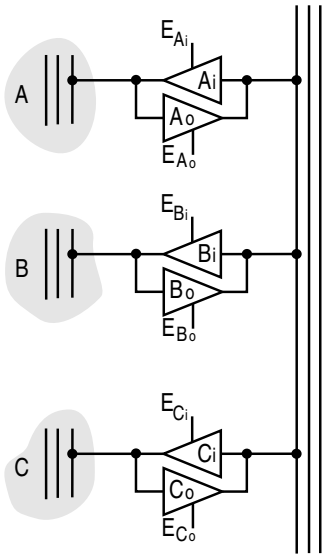
The Tri-State



E	X	F
0	0	–
0	1	–
1	0	0
1	1	1

- E is Circuit Enabled
- Enabled:
  Direct connection between input (X) and output (F)
- Disabled:
  Input *not* connected to output
  ⇒ Not included in fan in/out
  ⇒ No drain on power
  ⇒ Can connect one input to any number of disabled circuits
- Used to control access to a data highway or *bus*

The Bus



- Tri-States connect functional units (registers, counters, *etc*)
- Used to control access to a data path (also known as a *bus*)
- Devices A, B, and C connected to a system bus via tri-states
- Device A is connected only when the Enable signal (E_{AI} or E_{AO}) is high
  Not connected when Enable is low
- Data flows along the path

Programmable Logic Array

- Matrix of *and* and *or* gates
- Uses high voltage to make a connection
- Connections made by shorting matrix
- Programmed in high level using VHDL
  a C like language
- Erasable
- Reconfigurable Computing
  Rebuild the PLA on the fly
- Now over 1,000,000 Gates on a single Programmable Logic Array

Memory Technology

- Memory Cell
  Smallest unit of storage (1 bit)
- Access time
  Time taken to read/write memory
- Random-Access Memory (RAM)
  Access time to any cell is constant
- Serial- or Sequential-Access
  Access time is dependent on cell location
  Device accesses cells in sequence
- Volatile Memory
  Forgets contents when power removed
- Read-Only Memory (ROM)
  Can't modify contents (normally)
  Non-volatile by definition
  Normally Random-Access

Memory Hierarchy

Criteria:

Speed	(very low access time)
Small	(physically small)
Power	(low power consumption)
Robust	(extreme conditions)
Cost	(as low as possible)

1	Registers	Static memory on CPU
2	Cache (Level 1)	Static memory on CPU
3	Cache (Level 2)	Fast off CPU static memory
4	Main Memory	Off CPU static / dynamic memory
5	Magnetic Disk	Large storage, slow access
6	Optical Disk	Very large, but slow (ie., DVD)
7	Magnetic Tape	V. Large, V. Slow, V. Cheep

Semiconductor Memory

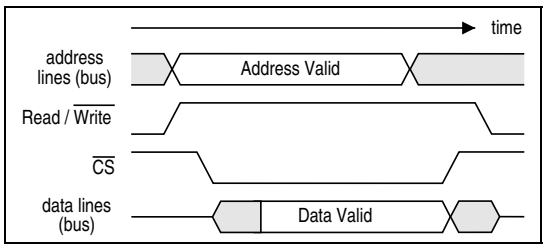
- Measured as *size* $\times$ *width*
  32Meg $\times$ 1 bit or 4Meg $\times$ 8 bits
- Control Lines:

$\overline{\text{CS}}$	*Chip Select*: Which chip to use
Read / $\overline{\text{Write}}$	Access type
A_0–A_n	*Address*: Location within chip
D_0–D_7	*Data*: value being transferred
Refresh	Dynamic refresh

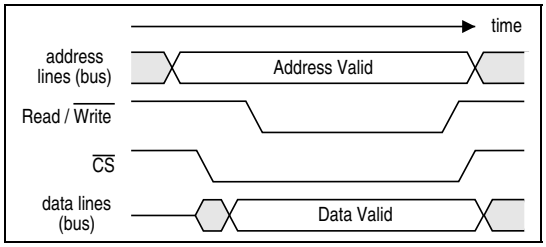
- Erasable Programmable Read-Only Memory (EPROM)
  Uses large current to enable write
  Non-volatile, long access time
  EEPROM requires 12v to erase or program

Accessing Memory

Read



Write



Sex

- Describes which bit of a byte is transmitted first:

  Big-Endian: Most Significant bit (Bit-7) first
  Used by Motorola, Networks and most RISC chips

  Little-Endian: Least Significant bit (Bit-0) first
  Used by Intel

- All hardware in a system must be of the same type, endian-neutral, or it must be converted
- Name comes from Swift's *Gulliver's Travels* where the Lilliputians go to war over which end a soft-boiled egg should be opened, big- or little-end ?

Parity Bit

- Extra bit added to a byte or word to reveal errors
- Count the number of 1 bits in the byte / word
- For even parity, set parity bit so that there are an even number of 1-bits, including parity bit
- Odd parity, parity bit for odd number of 1-bits
- A single parity bit is limited:

  ⇒ Can only reveal single bit errors
  parity same if even number of wrong bits

  ⇒ Can not tell which bit is wrong
  error detection and correction required

Types of Memory

SDRAM	Synchronous Dynamic Random Access Memory DRAM with separate clock for Burst mode access
DDR-RAM	Double Data Rate Random Access Memory Clocked on rising- and falling-edge, twice as fast?
EDO RAM	Extended Data Out DRAM Extra next-byte signal to reduce address decoding
SIMM	Single In-line Memory Module Small circuit board with integrated RAM chips
DIMM	Dual In-line Memory Module Two SIMMS back to back on one circuit board
SO-DIMM	Small Outline DIMM A DIMM on a smaller board, used in laptops